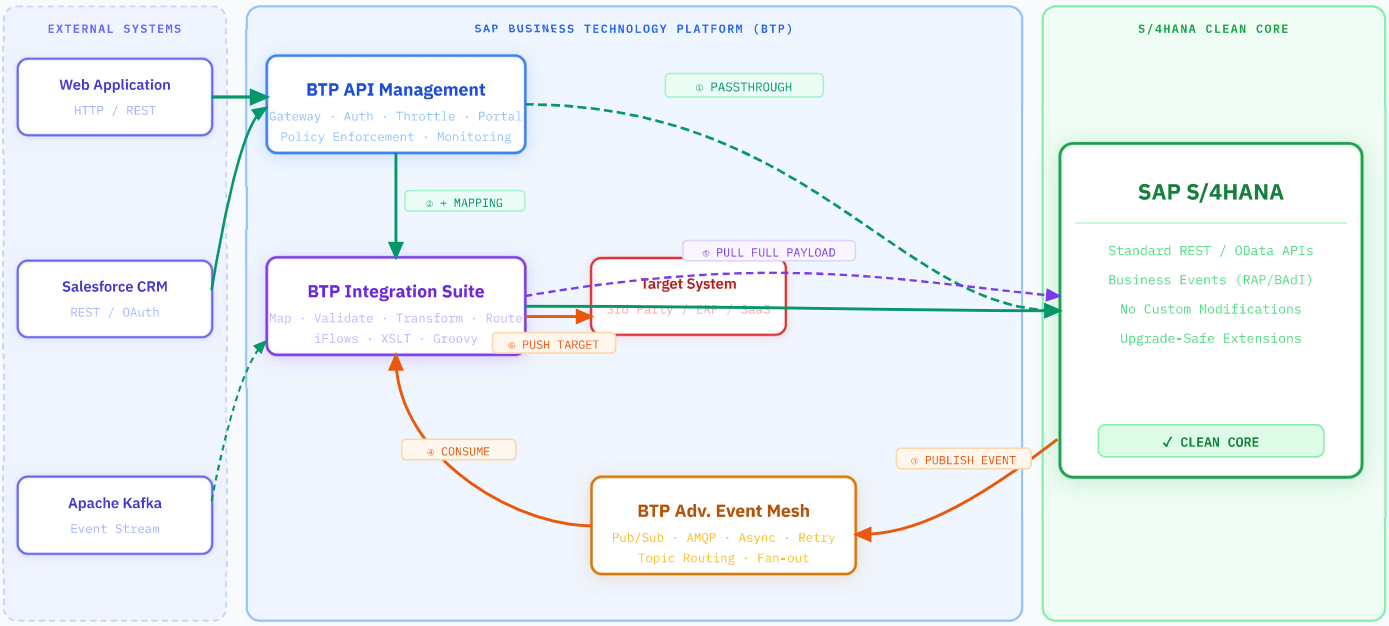
SAP S/4HANA — CLEAN CORE INTEGRATION ARCHITECTURE

BTP API MANAGEMENT · INTEGRATION SUITE · ADVANCED EVENT MESH



Inbound Flow (REST/API) · Passthrough (no mapping) · Outbound Event-Driven · Internal Callback (Pull Payload)

⬇ INBOUND INTEGRATION

→ All inbound calls are funnelled through **BTP API Management** — enforcing auth, throttling, and policy at the edge

→ **Passthrough** ①: If no payload transformation is needed, API Mgmt forwards the call directly to S/4HANA standard APIs

→ **Mapping required** ②: API Mgmt routes to BTP Integration Suite, which transforms & enriches the payload before calling S/4HANA

→ Source systems (Web Apps, Salesforce, Kafka) all enter through the same governed gateway — no direct back-door calls

⬆ OUTBOUND INTEGRATION

→ S/4HANA raises a **Business Event** ③ via RAP / BAdI when an object changes — no custom ABAP modifications

→ **BTP AEM** buffers the event asynchronously, decoupling SAP from downstream latency and availability

→ BTP IS **subscribes** ④ to the AEM topic and triggers an iFlow on event arrival

→ IS **calls back S/4HANA** ⑤ via REST API to pull the full business object payload (events carry only key IDs)

→ IS performs **mapping, validation & business logic**, then pushes the enriched message to the target system ⑥

✦ ARCHITECTURE DESIGN PRINCIPLES

→ **Clean Core**: Zero modifications to S/4HANA — all integration logic lives in BTP layers, ensuring upgrade safety

→ **AEM as decoupler**: Absorbs event bursts, enables fan-out to multiple consumers, and guarantees at-least-once delivery with replay

⚙ COMPONENT SUMMARY

→ **BTP API Management** — Inbound gateway, OAuth 2.0, rate limiting, developer portal & analytics

→ **BTP Integration Suite (CPI)** — iFlow orchestration, XSLT/Groovy mapping, message validation, content-based routing

→ **Single inbound entry point**: API Management provides unified observability, versioning, and developer self-service

→ **IS as orchestrator**: All mapping, enrichment, and routing logic centralised in iFlows — no logic scattered across adapters

→ **BTP Advanced Event Mesh** — AMQP/MQTT/REST event broker, topic-based routing, guaranteed delivery, event replay

→ **SAP S/4HANA** — Exposes standard OData/REST APIs, emits RAP business events, BAdI side-effects only